Ogbonna Arinze

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SUMMARY

Senior Full-Stack Software Engineer with 8 years of professional experience building production-grade web applications using React, Next.js, and TypeScript across healthcare, fintech, civic tech, and enterprise environments. Deep expertise in full-stack Next.js development, including server-side rendering, API route design, and clear client/server architectural boundaries. Proven track record of building reusable component libraries, orchestrating monorepo architectures with Nx and pnpm, and driving code quality through structured code reviews, automated testing (Playwright, Jest), and CI/CD standardization.

CORE COMPETENCIES

- **Full-Stack Architecture:** Next.js (SSR/SSG/ISR, API routes, server components), React, TypeScript, monorepos (Nx, pnpm workspaces), and modular architecture.
- **Frontend Development:** Scalable component systems, design systems (Radix UI), Redux, React Query, Zustand, Tailwind CSS, and performance optimization.
- **Backend & API:** NestJS, NodeJS REST APIs, PostgreSQL, Prisma, MongoDB, type-safe API integrations, JWT authentication, and RBAC.
- **Testing & Quality:** Playwright E2E, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, code reviews, and CI-driven QA.
- **Cloud & Infrastructure:** Docker, GitHub Actions, CI/CD pipelines, AWS (CloudFront, S3), Vercel, Nx monorepo orchestration, and Traefik.
- **Leadership:** Technical architecture reviews, mentorship, Agile/Scrum, TDD/BDD, and collaborative development practices.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Co-developed a full-stack **Next.js** chat widget powered by a Retrieval-Augmented Generation (RAG) system using OpenAI GPT models, **LangChain**, Vercel AI SDK, and Chroma vector database, implementing both client-side conversational UI and server-side data retrieval logic with clear architectural boundaries.
- Spearheaded the redesign of the main clinical trial data product, migrating from a Vue.js web application to a **React** Chrome extension with **TypeScript**, optimizing workflows for clinical site integrations and enhancing adoption.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, building shared components, services, and utilities that improved UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Implemented **JWT-based authentication** and **role-based access control** in Next.js with TypeScript and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering, collaborating with backend engineers on API contracts.
- Optimized application performance by implementing **lazy loading, code splitting**, dynamic imports, and CDN asset delivery, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components with TypeScript and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** with GitHub Actions, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Conducted structured **code reviews** to maintain quality and architectural consistency, and mentored a team of two junior developers (interns), fostering a culture of technical excellence.

Descasio Lagos, Nigeria (Hybrid)
Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of a scalable **React** frontend architecture with **TypeScript**, achieving a **25% boost in user engagement** and collaborating with software architects to accelerate development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports and improved code reliability.
- Utilized **Storybook** for documenting reusable React components, streamlining change management and supporting agile development practices with precise user story planning.
- Established engineering standards for code quality using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.

- Mentored junior engineers and conducted code reviews on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, collaborating closely with the business team to ensure technical solutions met business needs.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and code review feedback.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications** with **pnpm workspaces**: **NestJS** API backend, **Next.js** citizen-facing web app, Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, building reusable components, services, and utilities shared across all applications.
- Built the **Next.js** citizen-facing frontend with **React 19**, **Tailwind CSS v4**, and **Recharts**, featuring 22 interactive chart types auto-generated from AI responses, dark mode, URL state management with nuqs, responsive mobile layouts, and **server-side rendering** for SEO-optimized public conversation pages.
- Designed the **PostgreSQL 16** database schema with **Prisma v7** for **type-safe** database access across the NestJS backend, including pgvector extension for vector storage, and wrote custom migration tooling that filters out external tables from Prisma's drift detection.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, token-budgeted context management, and phone OTP authentication with session cookies, defining clear boundaries between client and server responsibilities.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and **Traefik** for reverse proxy routing.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*